Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [14CE64]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

ESTIMATING COSTING & SPECIFICATIONS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Part-A is compulsory which carries 28 marks. Answer any **THREE** full questions from Part-B which carries 14 marks each.
2. Missing data may be suitably assumed.
3. Follow the written dimension do not scale the drawing.

PART - A

1. The plan and section of one roomed building is shown in the figure-1.Prepare quantity estimate for the following items of work using long wall and short wall method. **28**

   i. Earth work excavation in foundation.

   ii. Earth work in plinth filling

   iii. Lime concrete in foundation

   iv. First class brick work using cement mortar (1:6) for foundation and plinth.

   v. 2.5cm thick damp proof course.

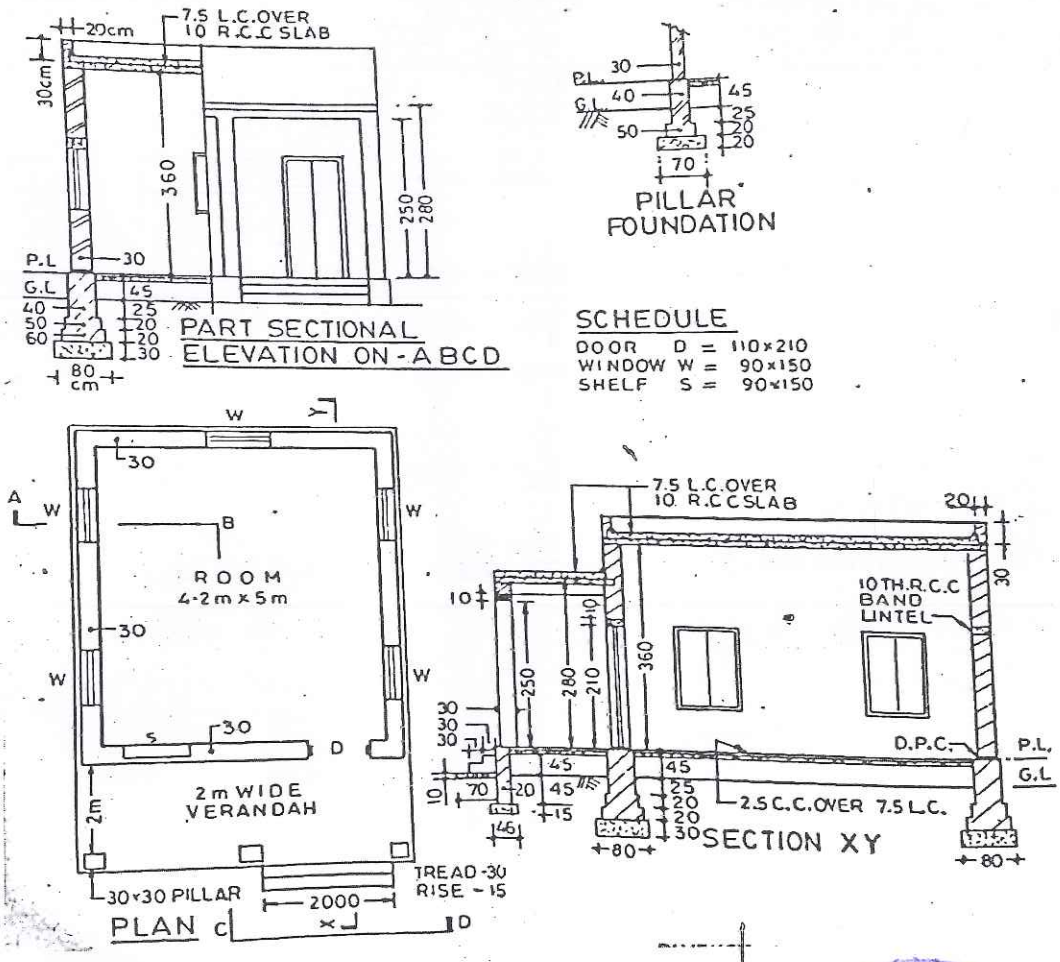

Figure.1

Note: All dimensions are in cm

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PART - B

2. A reinforced cement concrete (proportion 1:2:4) rectangular beam 20cm wide 30cm deep 3.0 meter overall length is reinforced with tor steel bars 3 numbers 16mm dia (weight =1.58kg/m) two outer bars straight and L-hooked at ends and the inner bar bent up at450 at appropriate places with L-hooked at ends. At top two outer hanger bars are 10mm in dia (weight= 0.62kg/m) straight and L-hooked at ends. Stirrups are 6mm in dia, MS bar (weight 0.22kg/m and spaced at 20cm center to center. All concrete cover 2.5cm. **14**

3. Estimate the quantity and cost of earth work for a road between two stations A to B with the following data. Width of road is 10meter at formation surface and side slope 2:1.Rate for earthwork in banking and cutting may be taken as Rs.10.00per cubic meter including a lead up to 150meter with a condition that portion of earth work available from cutting is to be utilized for banking within the same lead of 150meter with a condition that portion of earth available from cutting is to be utilized for banking within the same lead of 150 meter. The data of field book for the portion of road are as below adopt mid section method. **14**

Chainage	0	1	2	3	4	5	6
RL.	123.90	125.00	124.60	122.90	121.60	121.00	120.40
Formation Level	123.20	123.60	124.00	123.60	123.20	122.80	122.40

(Note: One Chain = 30Meter)

4. Prepare a detailed estimate of RCC cantilever retaining wall for a length of 30 meter from the **fig.2**, weight of 10mm dia bars = 0.62kg/m,12mm dia bars = 0.89 kg/m, 16mm diameter bars= 1.58kg/m and 20mm dia bars weight = 2.47Kg/m.Adopt the following rates to prepare the estimate.

   i. RCC work (1:2:4) including centring and shuttering but excluding reinforcement at Rs.1400 per cubic meter.

   ii. MS reinforcement including cutting, hooking etc at Rs.900 per quintal.

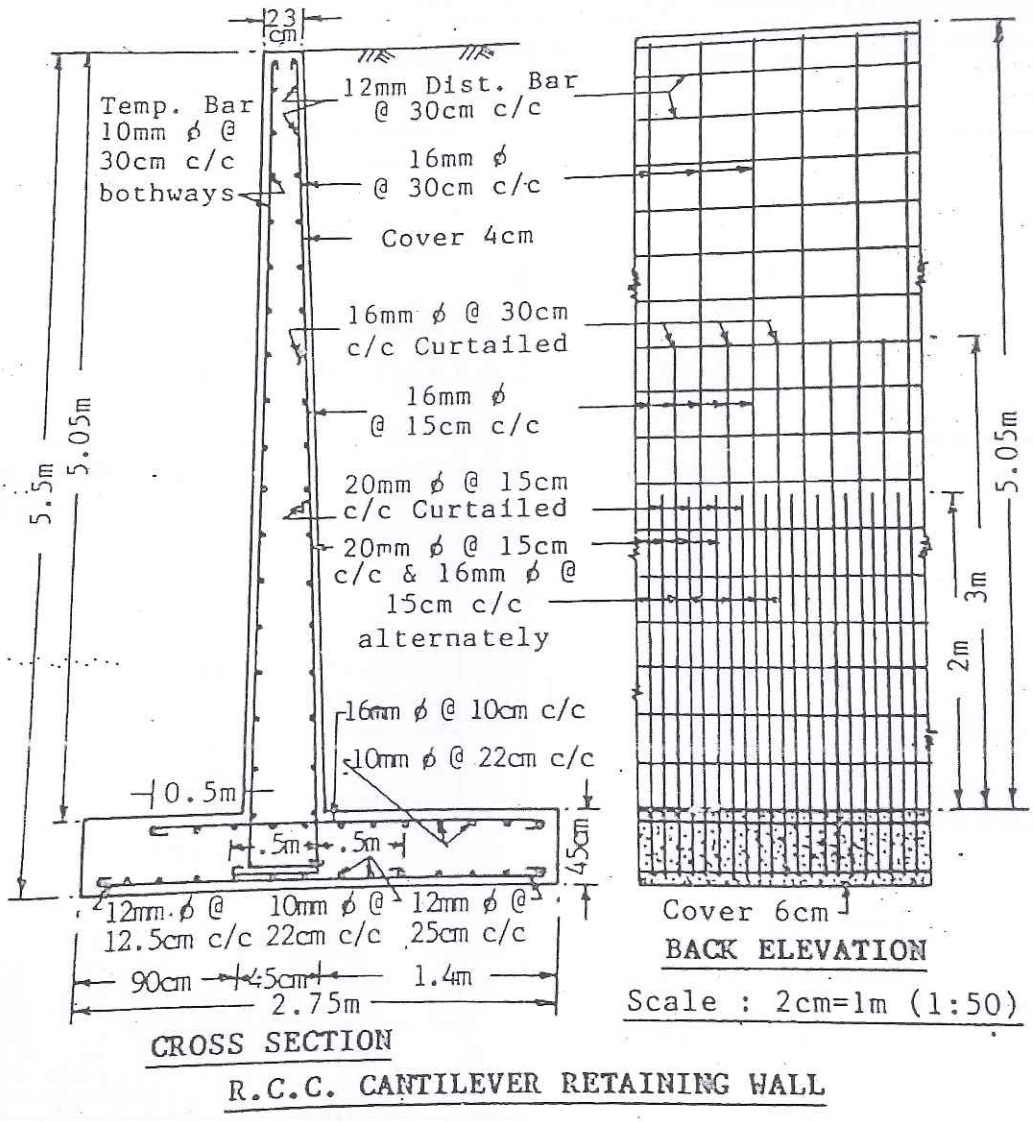

Figure-2

5. Estimate the quantity of timber required to construct a panelled door shutter as shown in the **figure-3,** frame includes and also prepare an analysis of rate for the shuttering only. All timber works shall be of first class Indian Teak Wood. Assume reasonable Market rates for your locality. **14**

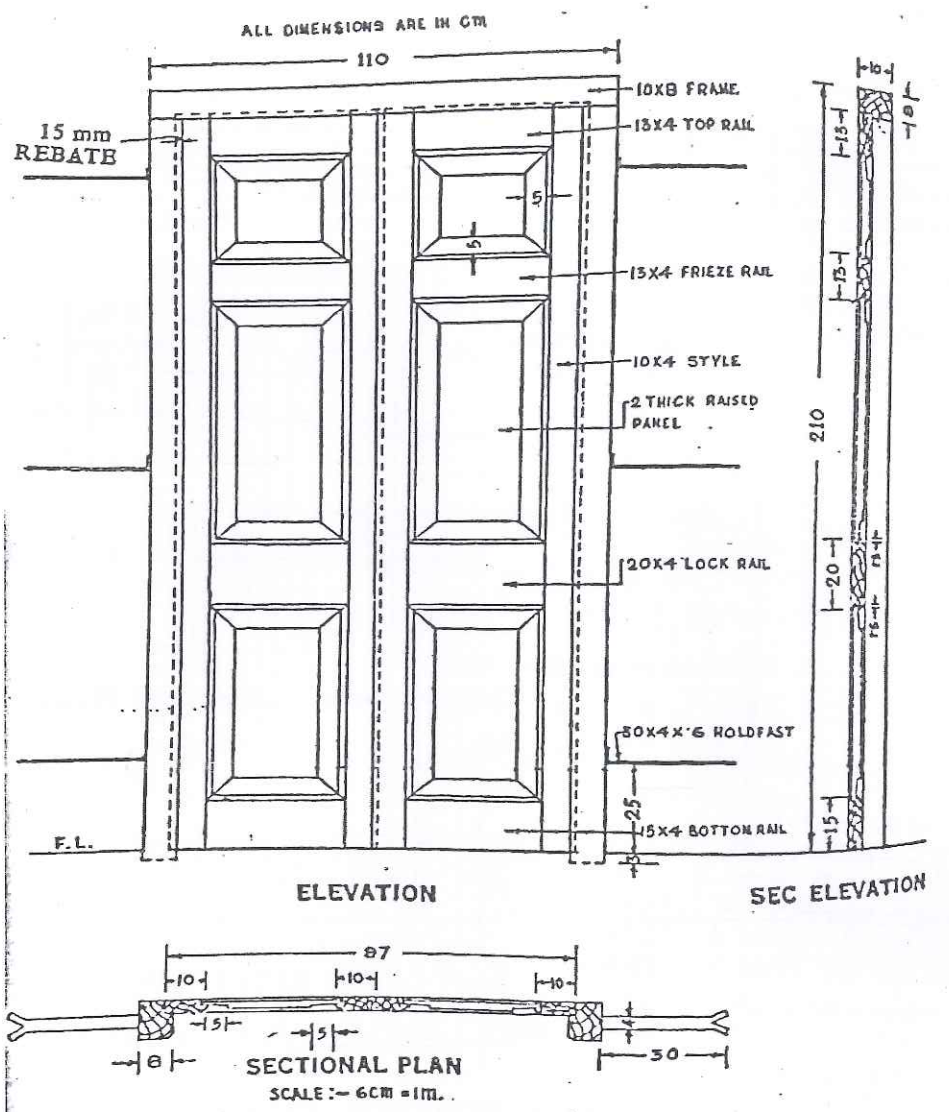

FIG. 8-4
PANELLED DOOR SHUTTER WITH FRAME

Figure-3

6. Explain the necessity of specifications. **07**

Explain the principles adopted in writing specifications. **07**

7. a. Mention the types of contracts and explain any three. **07**

b. Differentiate between security deposit and retention money. **07**